Databricks AutoML

Databricks refers to its AutoML solution as a **glass box**. This terminology highlights the fact that there is nothing hidden from the user. This feature in the Data Intelligence Platform leverages an open source library, Hyperopt, in conjunction with Spark for hyperparameter tuning. It intelligently explores different model types in addition to optimizing the parameters in a distributed fashion. The use of Hyperopt allows each run within the AutoML experiment to inform the next run, reducing the overall number of runs needed to reach an optimal solution compared to a grid search. Each run in the experiment has an associated notebook with the code for the model. This method increases productivity, reduces unnecessary computing, and lets scientists perform experiments instead of writing boilerplate code. Once AutoML has converged on the algorithmically optimal solution, there is a "best notebook" for the best scoring model. We'll expand on AutoML in several chapters throughout this book.

Reusability and reproducibility

As data scientists, transparency is especially important. We do not trust black box models. How do you use them without understanding them? A model is only as good as the data going in. In addition to not trusting the models, black boxes create concerns about our research's reproducibility and model drivers' explainability.

When we create a model, who does it belong to? Can we get access to it? Can we tweak, test, and, most importantly, reuse it? The amount of time put into the model's creation is not negligible. Databricks AutoML gives you everything to explain, reproduce, and reuse the models it creates. In fact, you can take the model code or model object and run it on a laptop or wherever. This open source, glass-box, reproducible, and reusable methodology is our kind of open.

Open file formats give you flexibility

Flexibility is also an essential aspect of the Databricks platform, so let's dive into the file format Delta, an open source project that makes it easy to adapt to many different use cases. For those familiar with Parquet, you can think of Delta as Parquet-plus – Delta files are Parquet files with a transaction log. The transaction log is a game changer. The increased reliability and optimizations make Delta the foundation of Databricks' lakehouse architecture. The data lake side of the lakehouse is vital to data science, streaming, and unstructured and semi-structured data formats. Delta has also made the warehouse side possible. There are entire books on Delta; see the *Further reading* section for some examples. We are focusing on the fact that it is an open file format with key features that support building data products.